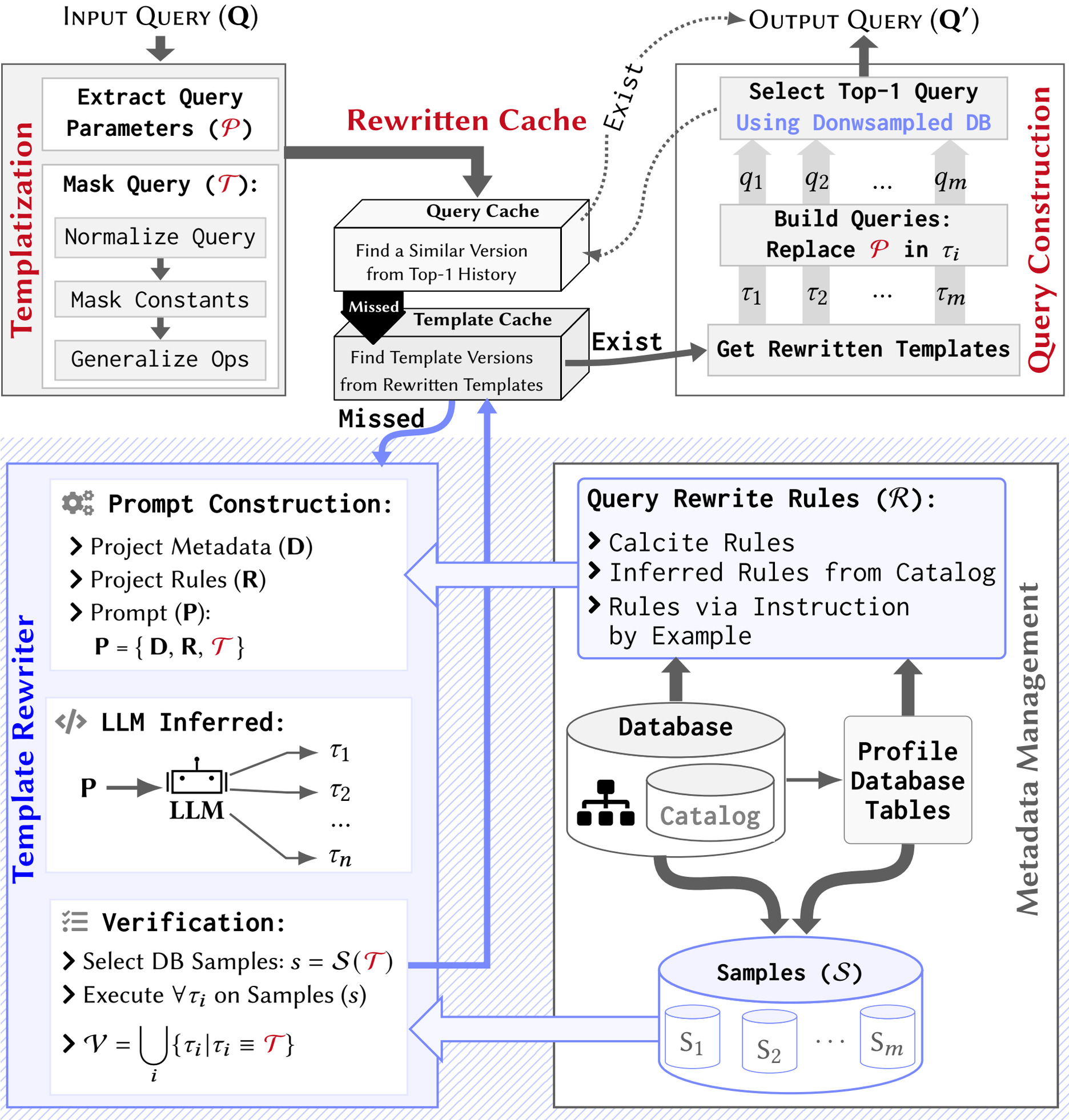

Input Query (Q)
Output Query (Q′)
Templatization
Extract Query Parameters (𝒫)
Mask Query (𝒯):
Normalize Query
Mask Constants
Generalize Ops
Rewritten Cache
Exist
Query Cache
Find a Similar Version from Top-1 History
Missed
Template Cache
Find Template Versions from Rewritten Templates
Exist
Missed
Select Top-1 Query Using Donwsampled DB
q1 q2 ... qm
Build Queries: Replace 𝒫 in τi
τ1 τ2 ... τm
Get Rewritten Templates
Query Construction
Template Rewriter
Prompt Construction:
Project Metadata (D)
Project Rules (R)
Prompt (P):
P = { D, R, 𝒯 }
LLM Inferred:
P
LLM
τ1
τ2
...
τn
Verification:
Select DB Samples: s = 𝒮(𝒯)
Execute ∀τi on Samples (s)
𝒱 = ⋃i {τi | τi ≡ 𝒯}
Query Rewrite Rules (ℛ):
Calcite Rules
Inferred Rules from Catalog
Rules via Instruction by Example
Database
Catalog
Profile Database Tables
Samples (𝒮)
S1 S2 ... Sm
Metadata Management